

4. Diagnostic Communications

4.1 Diagnostic Interface Protocols

The final tester utilises two diagnostic protocols over a 500 kb/s CAN bus interface for all communication to the Barra BEM. They are Ford Diagnostic (FD) and CAN Calibration Protocol (CCP).

FD commands are used to establish the CCP interface and read the software version number during the final test. CCP is used perform all functional tests on the BEM.

The following message identifiers are used to signify which protocol is being transmitted to the BEM:

Protocol	Request Identifier	Reply Identifier
FD	726	72E
CCP	706	712

Table 1 - Protocol Message Identifiers

4.2 Establishing CCP diagnostic communication

To establish, maintain connection and unlock the BEM for diagnostic commands use the following message.

Message	Protocol	Transmit	Receive
Enter CCP Mode	FD	07 2F F2 A9 07 97 6A 65	04 6F F2 A9 00 00 00 00 (Entered CCP Mode)
			03 7F 2F 12 00 00 00 00 (Unsuccessful)
Connect to BEM	CCP	01 01 BE BE 00 00 00 00	FF 00 01 00 00 00 00 00 (Connected)
			FF 32 01 00 00 00 00 00 (Invalid station address)
Get Security Seed*	CCP	12 02 02 00 00 00 00 00	FF 00 02 01 K1 K2 K3 K4 (Seed data = K1,K2,K3,K4)
			FF 00 02 00 K1 K2 K3 K4 (BEM already unlocked)
Unlock BEM	CCP	13 03 R1 R2 R3 R4 00 00	FF 00 03 02 00 00 00 00 (Successful unlock)
			FF 35 03 00 00 00 00 00 (Unsuccessful - incorrect seed)
Disconnect BEM	CCP	07 0A 01 00 BE BE 00 00	FF 00 0A 00 00 00 00 00 (Successful disconnect)
			FF 32 0A 00 00 00 00 00 (Out of Range)
Tester Present (with response)	FD	02 3E 01 00 00 00 00 00	01 7E 00 00 00 00 00 00 (Successful)
Tester Present (without response)	FD	02 3E 02 00 00 00 00 00	No Response

* See appendix for calculating R1 R2 R3 R4 from K1 K2 K3 K4

Note that the BEM diagnostic mode will time out after 5 seconds unless a tester present message is sent.

4.3 Block Download/Upload (Writing/Reading to/from EEPROM)

Message	Protocol	Transmit	Receive
Set MTA	CCP	02 04 00 00 ww xx yy zz	FF 00 04 00 00 00 00 00 (MTA set)
			FF 35 04 00 00 00 00 00 (Access locked)
			FF 32 04 00 00 00 00 00 (Out of range/Protected Area)
Download to EEPROM	CCP	03 05 ss uu uu uu uu 00	FF 00 05 00 00 00 40 04 (Data successfully downloaded)
			FF RR 05 00 00 00 00 00 (Unsuccessful)
Upload from EEPROM	CCP	04 06 ss 00 00 00 00 00	FF 00 06 uu uu uu uu uu (Data requested)
			FF RR 06 00 00 00 00 00 (Unsuccessful)
Short Upload From EEPROM	CCP	0F 07 ss 00 ww xx yy zz	FF 00 07 uu uu uu uu uu (Data requested)
			FF RR 07 00 00 00 00 00 (Unsuccessful)

wwwxyyzz = Memory transfer address

ss = Data size

uu = Data to be transferred

RR=

- 10 (Processor Busy)
- 22 (Initialise MAO)
- 32 (Parameter out of range)
- 35 (Access Locked)